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GTC Law Group PC & Affiliates
One University Ave., Ste. 302B
Westwood, MA 02090

EXAMINER	
ALGIBHAH, HAMZA N	

ART UNIT	PAPER NUMBER
2441	

NOTIFICATION DATE	DELIVERY MODE
09/25/2025	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

uspatents@gtclawgroup.com

Office Action Summary	Application No. 18/743,848	Applicant(s) Fang et al.	
	Examiner HAMZA N ALGIBHAH	Art Unit 2441	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 09/12/2025.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-2,4-9,11-16 and 18-21 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-2,4-9,11-16 and 18-21 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 06/14/2024 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

Detailed Action

1. ***Claims* 1-2, 4-9, 11-16 and 18-21** are pending.

2. ***Claims* 1-2, 4-9, 11-16 and 18-21 *are rejected.***

Double Patenting

The nonstatutory double patenting rejection is based on a judicially created doctrine grounded in public policy (a policy reflected in the statute) so as to prevent the unjustified or improper timewise extension of the "right to exclude" granted by a patent and to prevent possible harassment by multiple assignees. A nonstatutory double patenting rejection is appropriate where the conflicting claims are not identical, but at least one examined application claim is not patentably distinct from the reference claim(s) because the examined application claim is either anticipated by, or would have been obvious over, the reference claim(s). See, e.g., *In re Berg*, 140 F.3d 1428, 46 USPQ2d 1226 (Fed. Cir. 1998); *In re Goodman*, 11 F.3d 1046, 29 USPQ2d 2010 (Fed. Cir. 1993); *In re Longi*, 759 F.2d 887, 225 USPQ 645 (Fed. Cir. 1985); *In re Van Ornum*, 686 F.2d 937, 214 USPQ 761 (CCPA 1982); *In re Vogel*, 422 F.2d 438, 164 USPQ 619 (CCPA 1970); *In re Thorington*, 418 F.2d 528, 163 USPQ 644 (CCPA 1969).

A timely filed terminal disclaimer in compliance with 37 CFR 1.321(c) or 1.321(d) may be used to overcome an actual or provisional rejection based on nonstatutory double patenting provided the reference application or patent either is shown to be commonly owned with the examined application, or claims an invention made as a result of activities undertaken within the scope of a joint research agreement. See MPEP § 717.02 for applications subject to examination under the first inventor to file provisions of the AIA as explained in MPEP § 2159. See MPEP § 2146 *et seq.* for applications not subject to examination under the first inventor to file provisions of the AIA. A terminal disclaimer must be signed in compliance with 37 CFR 1.321(b).

The filing of a terminal disclaimer by itself is not a complete reply to a nonstatutory double patenting (NSDP) rejection. A complete reply requires that the terminal disclaimer be accompanied by a reply requesting reconsideration of the prior Office action. Even where the NSDP rejection is provisional the reply must be complete. See MPEP § 804, subsection I.B.1. For a reply to a non-final Office action, see 37 CFR 1.111(a). For a reply to final Office action, see 37 CFR 1.113(c). A request for reconsideration while not provided for in 37 CFR 1.113(c) may be filed after final for consideration. See MPEP §§ 706.07(e) and 714.13.

The USPTO Internet website contains terminal disclaimer forms which may be used. Please visit www.uspto.gov/patent/patents-forms. The actual filing date of the application in which the form is filed determines what form (e.g., PTO/SB/25, PTO/SB/26, PTO/AIA/25, or PTO/AIA/26) should be used. A web-based eTerminal Disclaimer may be filled out completely online using web-screens. An eTerminal Disclaimer that meets all requirements is auto-processed and approved immediately upon submission. For more information about eTerminal Disclaimers, refer to www.uspto.gov/patents/apply/applying-online/eterminal-disclaimer.

Claims 1-20 are provisionally rejected on the ground of nonstatutory double patenting as being unpatentable over claims 1-8 of Patent No. 12,103,479. Although the claims at issue are not identical, they are not patentably distinct from each other because claim 1 of the instant application states "automation description" instead of "task description" stated on claim 1 of the co-pending application. Since an "automation description" can be one possible type of a "task description", the claims are not patentably distinct from each other.

Instant claim	Patent 12,103,479 claim
1. An apparatus, comprising:	1. A system, comprising: **- a vehicle having a network comprising a plurality of end points;**

- a controller, comprising: - an automation definition circuit structured to interpret an automation description from an external device; - an automation management circuit structured to provide an automated action plan in response to the automation description; and - an automation execution circuit structured to provide an automation command in response to the automated action plan; wherein the automation command is structured to implement an automated vehicle response via an end point of a plurality of end points of a network of a vehicle.	- a controller, comprising: - an automation definition circuit structured to interpret a **task** description; - an automation management circuit structured to provide a **task** action plan in response to the **task** description; and - an automation execution circuit structured to provide a **task** execution command in response to the **task** action plan; wherein an end point of the plurality of end points is responsive to the **task** execution command to implement a **task** for the vehicle.

This is a provisional nonstatutory double patenting rejection because the patentably indistinct claims have not in fact been patented.

Claims 1-20 are provisionally rejected on the ground of nonstatutory double patenting as being unpatentable over claims 1-11 of Patent No. (not assigned yet) (reference application#: 18093964). Although the claims at issue are not identical, they are not patentably distinct from each other. Each of the limitation of the Patent claims in the above rejection is narrower than claim in the instant application. It would have been obvious to a person of ordinary skill in the art at the time the invention was made to omit elements wherein the remaining elements perform as before. A person of ordinary skill could have arrived at the present claims by omitting the details of the patented claims. See In re

Karlson (CCPA) 136 USPQ 184, decided January 16, 1963 ("Omission of element and its function in combination is obvious expedient if remaining elements perform same functions as before").

Instant claim	18093964 claim
1. An apparatus, comprising: - a controller, comprising: - an automation definition circuit structured to interpret an automation description from an external device; - an automation management circuit structured to provide an automated action plan in response to the automation description; and	1. A system, comprising: - a vehicle having a network comprising a plurality of end points; and - a controller, comprising: - an automation definition circuit structured to interpret an automation description generated by a device external to the vehicle, wherein the automation description is for at least one of: a speaker system of the vehicle; a seat heating system of the vehicle; a seat positioning system of the vehicle; an environmental management system of the vehicle; a lighting system of the vehicle; or an off-vehicle messaging system of the vehicle, and wherein the automation description includes an operation impact that specifies limits on utilization of resources of the vehicle - an automation management circuit structured to provide an automated action plan in response to the automation description, and to store the automated action plan as a data file on a data storage communicatively coupled to the controller; and

- an automation execution circuit structured to provide an automation command in response to the automated action plan;	- an automation execution circuit structured to provide an automation command in response to the automated action plan;
	- monitor the operation impact of the automation command for the limits on the utilization of resources; and generate a notification when the limits on the utilization have been exceeded, and
wherein the automation command is structured to implement an automated vehicle response via an end point of a plurality of end points of a network of a vehicle.	wherein an end point of the plurality of end points is responsive to the automation command to implement an automated vehicle response.

This is a provisional nonstatutory double patenting rejection because the patentably indistinct claims have not in fact been patented.

Claims 1-20 are provisionally rejected on the ground of nonstatutory double patenting as being unpatentable over claims 1-11 of copending Application No. 17833614 (reference application). Although the claims at issue are not identical, they are not patentably distinct from each other. Each of the limitation of the copending claims in the above rejection is narrower than claim in the instant application. It would have been obvious to a person of ordinary skill in the art at the time the invention was made to omit elements wherein the remaining elements perform as before. A person of ordinary skill could have arrived at the present claims by omitting the details of the patented claims. See In re Karlson (CCPA) 136 USPQ 184, decided January 16, 1963 ("Omission of element and its function in combination is obvious expedient if remaining elements perform same functions as before").

Instant claim	17833614 claim
1. An apparatus, comprising: - a controller, comprising: - an automation definition circuit structured to interpret an automation description from an external device; - an automation management circuit structured to provide an automated action plan in response to the automation description; and - an automation execution circuit structured to provide an automation command in response to the automated action plan; wherein the automation command is structured to implement an automated vehicle response via an end point of a plurality of end points of a network of a vehicle.	1. A system, comprising: - a vehicle having a network comprising a plurality of end points; - a controller, comprising: - an automation definition circuit structured to interpret an automation description; - an automation management circuit structured to provide an automated action plan in response to the automation description; and - an automation execution circuit structured to provide an automation command in response to the automated action plan; and wherein an end point of the plurality of end points comprises a sensor of the vehicle and wherein the sensor is responsive to the automation command to implement an automated vehicle response.

This is a provisional nonstatutory double patenting rejection because the patentably indistinct claims have not in fact been patented.

Claims 1-20 are provisionally rejected on the ground of nonstatutory double patenting as being unpatentable over claims 1-20 of copending Application No. 18743878 (reference application). Although the claims at issue are not identical, they are not patentably distinct from each other. Each of the

limitation of the copending claims in the above rejection is narrower than claim in the instant application. It would have been obvious to a person of ordinary skill in the art at the time the invention was made to omit elements wherein the remaining elements perform as before. A person of ordinary skill could have arrived at the present claims by omitting the details of the patented claims. See In re Karlson (CCPA) 136 USPQ 184, decided January 16, 1963 ("Omission of element and its function in combination is obvious expedient if remaining elements perform same functions as before").

18743878 claim	Instant claim
1. An apparatus, comprising:	1. A system, comprising: - a vehicle having a network comprising a plurality of end points; and
- a controller, comprising: - an automation definition circuit structured to interpret an automation description from an external device; - an automation management circuit structured to provide an automated action plan in response to the automation description; and	- a controller, comprising: - an automation definition circuit structured to interpret an automation description; - an automation management circuit structured to provide an automated action plan in response to the automation description, and to store the automated action plan as a data file on a data storage communicatively coupled to the controller; and
- an automation execution circuit structured to provide an automation command in response to the automated action plan; wherein the automation command is structured to implement an automated vehicle response via an end	- an automation execution circuit structured to provide an automation command in response to the automated action plan; wherein an end point of the plurality of end points is

point of a plurality of end points of a network of a vehicle.	responsive to the automation command to implement an automated vehicle response.

This is a provisional nonstatutory double patenting rejection because the patentably indistinct claims have not in fact been patented.

Claim Rejections - 35 USC § 103

1. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

2. The factual inquiries for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.

2. Ascertaining the differences between the prior art and the claims at issue.

3. Resolving the level of ordinary skill in the pertinent art.

4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

3. **Claim 1-2, 4-9, 11-16 and 18-21 are rejected under 35 U.S.C. 103 as being unpatentable over Muta et al. (Publication No. US 20190258252 A1; hereafter Muta) in view of Maluf et al. (Publication No. US 20190308589 A1; hereafter Maluf) and Onimaru, Hiroyuki (Publication No. US 20170147989 A1; hereafter Onimaru).**

4. Regarding to claim 1, Muta teaches **a system, comprising:**

a vehicle having a network comprising a plurality of end points; *(Fig. 2, "driving unit")*

a controller *(fig. 2, "control unit")***, comprising:**

- **an automation definition circuit structured to interpret an automation description;** *([Par. 0044], "The operation plan generating unit **1031** acquires the operation command from the management server **400**, and the operation plan generating unit **1031** generates the operation plan of itself. The information, which relates to the departure place and the destination given to the moving body **100**, is included in the operation command.")*

- **an automation management circuit structured to provide an automated action plan in response to the automation description,** *([Par. 0044], "The operation plan generating unit **1031** acquires the operation command from the management server **400**, and the operation plan generating unit **1031** generates the operation plan of itself. The information, which relates to the departure place and the destination given to the moving body **100**, is included in the operation command. Therefore, the operation plan generating unit **1031** calculates the moving route of the moving body **100**, and the operation plan generating unit **1031**generates the operation plan for moving along the moving route, on the basis of the destination given from the management server **400** and the position of the moving body **100** itself obtained from the position information acquiring unit **102**.")*

- ***provides an automated vehicle function*** *([Par. 0046], " The operation plan, which is generated by the operation plan generating unit 1031, is transmitted to the traveling control unit 1033 described later on."; [Par. 0048], "The traveling control unit 1033 generates the control command in order to control the autonomous traveling of the moving body 100 on the basis of the operation plan generated by the operation plan generating unit 1031, the environment data generated by the environment detecting unit 1032, and the position information of the moving body 100 acquired by the position information acquiring unit 102.")*

- **an automation execution circuit structured to provide an automation command in response to the automated action plan as a ~~data~~ parameter provided to an endpoint as a network communication;** *([Par. 0046], " The operation plan, which is generated by the operation plan generating unit 1031, is transmitted to the traveling control unit 1033 described later on."; [Par. 0048], "The traveling control unit 1033 generates the control command in order to control the autonomous traveling of the moving body 100 on the basis of the operation plan generated by the operation plan generating unit 1031, the environment data generated by the environment detecting unit 1032, and the position information of the moving body 100 acquired by the position information acquiring unit 102.")* **and**

- **wherein an end point of the plurality of end points is responsive to the ~~data~~ parameter of the automation command to implement an automated vehicle response.** *([Par. 0051], "The driving unit 104 is the means for allowing the moving body 100 to travel on the basis of the control command generated by the traveling control unit 1033. The driving unit 104 is constructed to include, for example, a motor for driving wheels provided for the moving body 100, an inverter, a brake, and a steering mechanism. For example, the motor and the brake are driven in accordance with the control command, and thus the autonomous traveling of the moving body 100 is realized.").*

Muta does not explicitly disclose that the automation command is provided **as a data parameter.** However, Maluf discloses that the automation command is provided **as a data parameter;** *([Par. 0016], "Serial networks* are another type of network, different from an IP network, typically forming a localized network in a given environment, such as for automotive or vehicular networks, industrial networks, entertainment system networks, and so on. For example, those skilled in the art will be familiar with the on-board diagnostics (OBD) protocol (a serial network which supports a vehicle's self-diagnostic and reporting capability, including the

upgraded "OBD II" protocol), the Controller Area Network (CAN) bus (or CANBUS) protocol (a message-based protocol to allow microcontrollers and devices to communicate with each other in applications without a host computer), and the MODBUS protocol (a serial communications protocol for use with programmable logic controllers, such as for remote terminal units (RTUs) in supervisory control and data acquisition (SCADA) systems). Unlike an IP-based network, which uses a shared and open addressing scheme, *a serial communication network* generally is based on localized and proprietary communication standards, *where commands or data are transmitted* based on localized device identifiers, such as parameter identifiers (PIDs), localized station addresses, and so on*")*

It would have been obvious to one of ordinary skill in the art before the effective filing date of the claim invention to modify Muta to incorporate the teaching of Maluf so that the command is provided as a data parameter because this would have allowed different vehicle components to communicate using serial communication network.

Muta teaches to generate traveling plan based on automation description as described above, but Muta and Maluf do not explicitly disclose **to store the automated action plan as a data file on a data storage communicatively coupled to the controller.** However, Onimaru teaches **to store the automated action plan as a data file on a data storage communicatively coupled to the controller;** *([Par. 0075], "the user information storage unit 111 stores various types of data (for example, a destination setting data file, action plan data, and the like described later) set and generated by the action planning system 100 for each user ID or each portable device ID (on-board navigation device ID).")*

It would have been obvious to one of ordinary skill in the art before the effective filing date of the claim invention to modify Muta and Maluf to incorporate the teaching of Onimaru. Storing the action plan would have been an obvious modification, as it enables the system to retrieve

the plan for later use or update it for future user demands. Moreover, by keeping a record of past action plans, the system can use this historical data to generate more effective action plans for future use.

Examiner Note: The components labeled as the 'automation definition circuit,' 'automation management circuit,' and 'automation execution circuit' are asserted to form integral parts of a unified controller. Breed 's system, as depicted in paragraph 0105 as an example, accomplishes the described functions through an 'n electronic control unit (ECU) ' Consequently, it is to be inferred that the ECU inherently encompasses all essential circuits required to execute the aforementioned functions.

As per claim 2, claim 1 is incorporated and the combination of Muta and Onimaru teaches the system of claim 1.

Onimaru further teaches **wherein the automation command comprises at least one data value determined from the data file.** *([Par. 0209], "the destination setting instruction unit 203 of the portable device 2 sets initial values for a moving means and a moving speed from the n-th destination to the (n+1)th destination for each n (1≦n≦N) when creating an action plan which involves a movement. By doing so, it is possible to adjust the action plan and support the execution thereof so that the user's desire is applied more satisfactorily)*

As per claim 4, claim 1 is incorporated and the combination of Muta and Onimaru teaches the system of claim 1.

Muta further teaches **wherein the automated action plan implements a conditional operation of the automation command.** *([Par. 0034], "The management server 400 gives the instruction to the moving body 100 to perform the control so that the working environment is adjusted on the basis of the work information in relation to the work to be executed by the user*

in the moving body 100. Further, the management server 400 gives the instruction to the moving body 100 to perform the control so that the moving body100 is moved to a predetermined position (destination) in accordance with the operation schedule."; [Par. 0044], "The operation plan generating unit 1031 acquires the operation command from the management server 400, and the operation plan generating unit 1031 generates the operation plan of itself. The information, which relates to the departure place and the destination given to the moving body 100, is included in the operation command." Wherein the indicated destination and operation scheduled in operation command read on the "conditional operation")

As per claim 5, claim 1 is incorporated and the combination of Muta and Onimaru teaches the system of claim 1.

Muta further teaches **wherein the automated action plan implements a branched operation of a plurality of automation commands, the plurality of automation commands including the automation command.** *([Par. 0048], "The traveling control unit 1033 generates the control command in order to control the autonomous traveling of the moving body 100 on the basis of the operation plan generated by the operation plan generating unit 1031, the environment data generated by the environment detecting unit 1032, and the position information of the moving body 100 acquired by the position information acquiring unit 102."; [Par. 0051], "The driving unit 104 is the means for allowing the moving body 100 to travel on the basis of the control command generated by the traveling control unit 1033. The driving unit 104 is constructed to include, for example, a motor for driving wheels provided for the moving body 100, an inverter, a brake, and a steering mechanism. For example, the motor and the brake are driven in accordance with the control command, and thus the autonomous traveling*

of the moving body 100 is realized." Where this is interpreted as the control signal transmitted to the driving control unit which implements motor, inverter, brake, steering mechanism for performing a plurality of automation command for controlling the autonomous driving corresponding to the traveling plan. Note that Examiner interprets the "automation commands" as "control signal" generated by the controller.)

As per claim 6, claim 1 is incorporated and the combination of Muta and Onimaru teaches the system of claim 1.

Muta further teaches **wherein the automated action plan implements a state machine operation including the automation command.** *([Par. 0048], "The traveling control unit 1033 generates the control command in order to control the autonomous traveling of the moving body 100 on the basis of the operation plan generated by the operation plan generating unit 1031, the environment data generated by the environment detecting unit 1032, and the position information of the moving body 100 acquired by the position information acquiring unit 102."; [Par. 0051], "The driving unit 104 is the means for allowing the moving body 100 to travel on the basis of the control command generated by the traveling control unit 1033. The driving unit 104 is constructed to include, for example, a motor for driving wheels provided for the moving body 100, an inverter, a brake, and a steering mechanism. For example, the motor and the brake are driven in accordance with the control command, and thus the autonomous traveling of the moving body 100 is realized." Where this is interpreted as the control signal transmitted to the driving control unit which implements motor, inverter, brake, steering mechanism for performing a plurality of automation command for controlling the autonomous driving corresponding to the traveling plan. The operation of the motor, inverter, brake and steering mechanism reads on the "state machine operation".)*

As per claim 7, claim 1 is incorporated and the combination of Muta and Onimaru teaches the system of claim 1.

Muta further teaches **wherein the automated action plan implements a cyclical operation including the automation command.** *([Par. 0048], "The traveling control unit 1033 generates the control command in order to control the autonomous traveling of the moving body 100 on the basis of the operation plan generated by the operation plan generating unit 1031, the environment data generated by the environment detecting unit 1032, and the position information of the moving body 100 acquired by the position information acquiring unit 102."; [Par. 0051], "The driving unit 104 is the means for allowing the moving body 100 to travel on the basis of the control command generated by the traveling control unit 1033. The driving unit 104 is constructed to include, for example, a motor for driving wheels provided for the moving body 100, an inverter, a brake, and a steering mechanism. For example, the motor and the brake are driven in accordance with the control command, and thus the autonomous traveling of the moving body 100 is realized."* ***Note:*** *the "cyclical operation" could be interpreted as a repeating behavior of one or more tasks or a repetitive driving task that an autonomous vehicle performs continuously. For this reason, controlling the steering, braking, and output of a motor are all key aspects of the cyclical operation in autonomous driving. In a cyclical operation, the autonomous vehicle would need to navigate the same route repeatedly, so it would need to make adjustments to steering, braking, and throttle based on the traveling plan, current driving conditions and the vehicle's speed and position. For example, if the vehicle is driving on a straight road, it may need to adjust the steering to maintain its position within the lane. If the vehicle is approaching a stop sign, it may need to apply the brakes to slow down and come to a stop. And if the vehicle needs to accelerate to match the speed of traffic, it may need to adjust the engine throttle to increase speed. Controlling these aspects of the vehicle's*

operation is essential to ensure safe and reliable autonomous driving in cyclical operations. Therefore, the operation of the autonomous vehicle according to the traveling plan reads on "cyclical operation")

***Claims* 8-9, 11-16 and 18-21 *are rejected under the same rationale as claims* 1-2 and 4-7.**

As per claim 21, claim 1 is incorporated and the combination of Muta and Onimaru teaches the system of claim 1.

Onimaru further teaches **wherein the data file provides an automated vehicle function that executes without changing at least one of software or firmware on the vehicle.** *([Par. 0075], "the user information storage unit 111 stores various types of data (for example, a destination setting data file, action plan data, and the like described later) set and generated by the action planning system 100 for each user ID or each portable device ID (on-board navigation device ID).")* Onimaru *destination setting data file, action plan data, and the like stored in the suer information storage unit does not change to at least one of software or firmware on the vehicle.*

Response to Arguments

Applicant's arguments filed 09/12/2025 have been fully considered but they are now moot in light of the new grounds of rejection.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to HAMZA N ALGIBHAH whose telephone number is (571)270-7212. The examiner can normally be reached 7:30 am - 3:30 pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Wing Chan can be reached at (571) 272-7493. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/HAMZA N ALGIBHAH/
Primary Examiner, Art Unit 2441